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Satellite Design & Integration

DAY 4

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OBC Exercises

Let's start coding the On Board Computer (OBC):

- Open a *new sketch* and save it as OBC_Exercises.

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OBC Exercises

Let's start coding the On Board Computer (OBC):

- Open a *new sketch* and save it as OBC_Exercises.
- Write an OBC program that displays the time elapsed (since the OBC is turned on) in seconds via serial monitor.

The output should look like:

TIME = 1s

TIME = 2s

TIME = 3s

...

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OBC Exercises

Let's start coding the On Board Computer (OBC):

- Open a *new sketch* and save it as OBC_Exercises.
- Write an OBC program that displays the time elapsed (since the OBC is turned on) in seconds via serial monitor.

The output should look like:

TIME = 1s

TIME = 2s

TIME = 3s

...

- Once done, include minutes in the display (remember that when we get to 59s we add a minute, resetting the seconds).

The output should look like:

TIME = 0m 25s

...

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OBC Exercises

- Request the time from the OBC through input (Serial.read) of the character 'T'. Arduino does not display anything via serial until requested (only one reading). The time counting must occur in the background, regardless of when I ask for 'T'.

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OBC Exercises

- Request the time from the OBC through input (Serial.read) of the character 'T'. Arduino does not display anything via serial until requested (only one reading). The time counting must occur in the background, regardless of when I ask for 'T'.
- Find a way to send the character 'R' and reset the time (not Arduino) so that the second time I request 'T', I will have a lower value than before.

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OBC Exercises - Analog Sensors

Below you will find the correct way to set up an analog sensor.

As you can see, in contrast to digital sensors, with analog sensors, **there's no need** to use the **pinMode(name, INPUT)** function..

```
int sensorPin = A0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int reading = analogRead(sensorPi
  Serial.print("Sensor Reading: ");
  Serial.println(reading);
  delay(1000);
}
```

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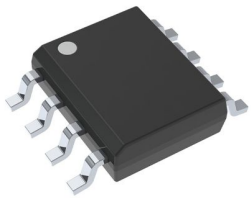
OBC Exercises

- The OBC has a temperature sensor, **LM35**, which is an analog sensor connected to pin A1.
- Look at the online datasheet of the LM35 to understand how it reads temperature and find a way to display it (char) in the serial monitor every 2 seconds.

The output should look like:

TEMPOBC = 26.2 deg



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TEXAS INSTRUMENTS

LM35

SNIS159H – AUGUST 1999 – REVISED DECEMBER 2017

LM35 Precision Centigrade Temperature Sensors

1 Features **3 Description**

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OBC Exercises

- The OBC has a temperature sensor, **LM35**, which is an analog sensor connected to pin A1.
- Look at the online datasheet of the LM35 to understand how it reads temperature and find a way to display it (as a float) in the serial monitor every 2 seconds.

The output should look like:

TEMPOBC = 26.2 deg

- Find a way to request the temperature using the string 'TEMPOBC' and display it on the serial monitor when requested.

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OBC Exercises

- The OBC has a temperature sensor, **LM35**, which is an analog sensor connected to pin A1.
- Look at the online datasheet of the LM35 to understand how it reads temperature and find a way to display it (as a float) in the serial monitor every 2 seconds.

The output should look like:

TEMPOBC = 26.2 deg

- Find a way to request the temperature using the string 'TEMPOBC' and display it on the serial monitor when requested.
- Add the the possibly of requesting “TIMEOBC”, “RTIMEOBC”,“TEMPOBC” separately in the same Arduino code, dedicating a function for each of them to be called in the void loop().

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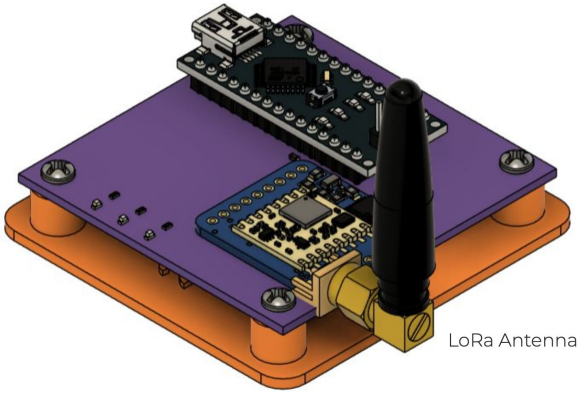
OBC/GS Exercises - LoRa

Now, let's shift our focus from the CubeSat to the ground station.

We will explore data transmission by uploading code exclusively to the ground station.

My ground station will serve as the receiver for all your ground station messages, as there is no cryptography involved.



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OBC/GS Exercises - LoRa

- Find the appropriate LoRa example (File > Examples > LoRa) for transmitting and open it.
- Always use the LoRa.print**ln**() function to send messages to the receiving ground station.
- In the example you will find a counter, remove it so that you send only one message (one every 5 seconds) - stick to this delay time.
- The correct frequency is 434E6.
- Send to my GS a message! Keep it short!

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OBC/GS Exercises - LoRa

Now, keep the Arduino connected to your ground station. Take your PC, divide your team, with some members staying inside the room and others going outside to move around the area, checking the range of communication. Use your smartphone to call your teammates and determine exactly when the communication stops. Ensure that the message you send includes your identification.

OBC/GS Exercises - LoRa

- The code from the previous exercise continuously sends the uploaded message on Arduino. Find a way to input a message in the serial monitor (using Serial.read) and ensure it is sent only once.

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OBC/GS Exercises - LoRa

- The code from the previous exercise continuously sends the uploaded message on Arduino. Find a way to input a message in the serial monitor (using Serial.read) and ensure it is sent only once.
- After completing the previous step, figure out how to include a timestamp along with the message.

The output should look like:
[0m 30s] your_message

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OBC Exercises

Now, let's configure the LoRa sender on the satellite, ensuring that the RBF is plugged in.

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OBC Exercises

Now, let's configure the LoRa sender on the satellite, ensuring that the RBF is plugged in. Let’s upload code to the **OBC ONLY**.

- Transmit the temperature from the OBC via radio, along with a brief message to identify yourself.

The output should look like:
'[NAME OF GROUP] | TEMPOBC = 20.4

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OBC Exercises

Now, let's configure the LoRa sender on the satellite, ensuring that the RBF is plugged in. Let’s upload code to the **OBC ONLY**.

- Transmit the temperature from the OBC via radio, along with a brief message to identify yourself.

The output should look like:
[NAME OF GROUP] | TEMPOBC = 20.4

- Now, figure out a way to send the temperature when inputting 'TEMPOBC' via serial. Implement the option to send also the time with 'TIME'.

The output should look like:
[NAME OF GROUP] | TEMPOBC = 20.4 | TIME = 0m 34s

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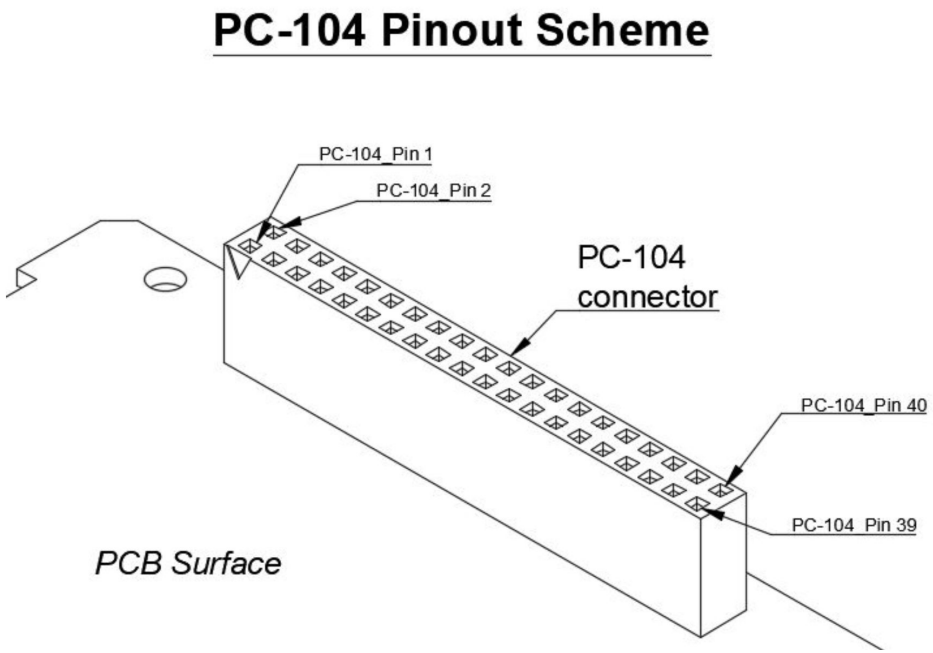
OBC Exercises

We can read sensors from other boards around the satellite using the industry-standard PC104 connector.



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OBC Exercises

PC-104 Connections

	Connections						
NOTES	EPS	OBC	ADCS	Pin Assignment	PC-104		Pin Assignment
Ground	✓	✓	✓	GND	1	2	NC
OUT of the solar charger	✓	✓	✓	VBUS	3	4	NC
Ground	✓	✓	✓	GND	5	6	NC
				NC	7	8	NC
				NC	9	10	NC
+5V bus from the OBC DC-DC converter	X	✓	X	OBC_5V	11	12	NC
				NC	13	14	NC
Battery Voltage (do not use it for power)	✓	✓	X	VBAT	15	16	NC
				NC	17	18	NC
ADCS reset signal from the OBC	X	✓	✓	ADCS_RESET	19	20	NC
				NC	21	22	NC
Temperature sensor on ADCS board	X	✓	✓	ADCS_TEMP	23	24	NC
Temperature sensor on EPS board	✓	✓	X	EPS_TEMP	25	26	NC
				NC	27	28	NC
ADCS +5V DC-DC ENABLE	X	✓	✓	ADCS_EN	29	30	NC
ADCS - OBC Serial port	X	✓	✓	RX_ADCS_TX_OBC	31	32	NC
ADCS - OBC Serial port	X	✓	✓	TX_ADCS_RX_OBC	33	34	NC
Ground	✓	✓	✓	GND	35	36	NC
OUT of the solar charger	✓	✓	✓	VBUS	37	38	NC
Ground	✓	✓	✓	GND	39	40	NC

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OBC Exercises

- Now, referring to the OBC schematics, locate the connection of the EPS temperature sensor on the OBC (no need to send it via LoRa). Write a code to read the temperature of the EPS - the output in your serial monitor should look like: 'TEMPEPS = 20.6.' The EPS utilizes the same temperature sensor as the OBC (LM35).

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OBC Exercises

- Now, referring to the OBC schematics, locate the connection of the EPS temperature sensor on the OBC (no need to send it via LoRa). Write a code to read the temperature of the EPS - the output in your serial monitor should look like: 'TEMPEPS = 20.6.' The EPS utilizes the same temperature sensor as the OBC (LM35).
- Figure out how to request 'TEMPOBC,' 'TEMPEPS,' and 'TIME' via serial so that Arduino provides the desired information when asked. Remember to create separate functions for each.

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OBC Exercises

- Now, referring to the OBC schematics, locate the connection of the EPS temperature sensor on the OBC (no need to send it via LoRa). Write a code to read the temperature of the EPS - the output in your serial monitor should look like: 'TEMPEPS = 20.6.' The EPS utilizes the same temperature sensor as the OBC (LM35).
- Figure out how to request 'TEMPOBC,' 'TEMPEPS,' and 'TIME' via serial so that Arduino provides the desired information when asked. Remember to create separate functions for each.
- Create a function, maxTemp(), that compares two temperatures. Invoke the function by entering 'TEMPCOMPARISON' via serial.

The output should should look like:
TEMPx is greater than TEMPy

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OBC Exercises

- Include in the function above the absolute difference between the temperatures.

The output should look like:

TEMPx is greater than TEMPy - Difference = 0.2 deg

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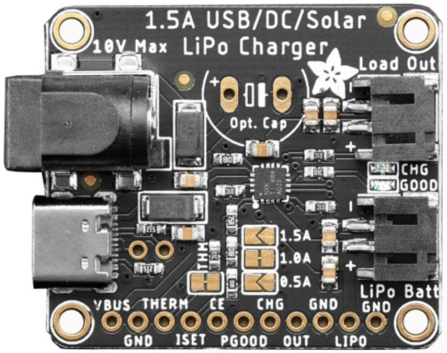
OBC Exercises

- Include in the function above the absolute difference between the temperatures.

The output should look like:

TEMPx is greater than TEMPy - Difference = 0.2 deg

- Refer to the OBC schematics to determine how to read the battery voltage (VBAT) using the **Adafruit 4755 solar charger**. Consult the datasheet online for accurate calculations of the battery voltage. Enable it to be called via serial by typing 'VBAT'.



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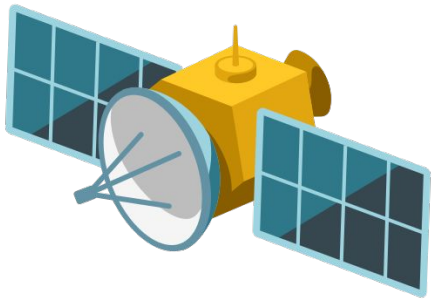

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